

Mark schemes

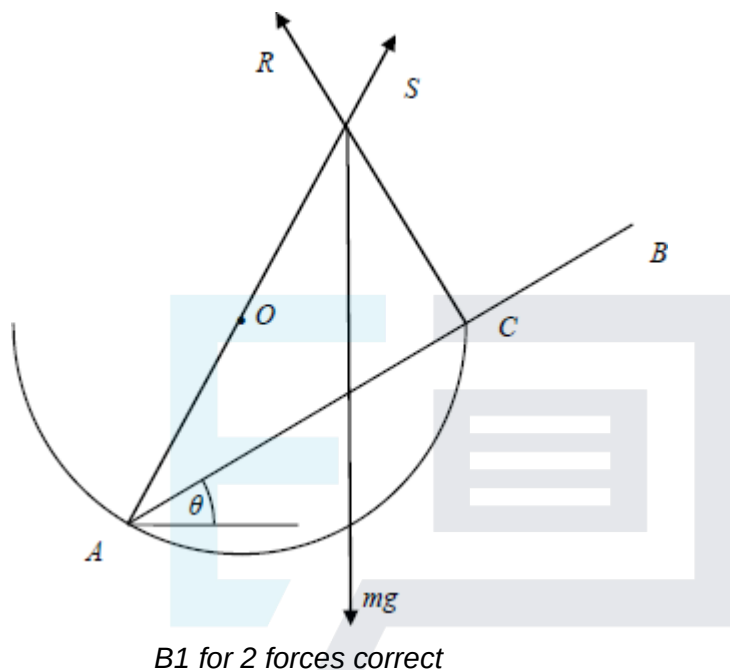
Q1.

- (a) Smooth, hence reaction is perpendicular to possible movement

E1

1

- (b)



B2

2

- (c) Resolving along the rod:

Or geometrically:

$$S \cos \theta = mg \sin \theta$$

three forces act through a point B1

M1 is for 2 or 3 terms; 1 term correct (could be horizontal force at C used)

Moment about C: $S 2a \cos \theta \cdot \sin \theta$

[forces act through point D]

M1A1

$$\text{Moment about C: } S 2a \cos \theta \cdot \sin \theta$$

$$= mg(2a \cos \theta - \frac{1}{2} l) \cos \theta$$

$$AD \cos 2\theta = \frac{l}{2} \cos \theta \quad \text{M1A1}$$

M1A1

$$4a.S \sin \theta = mg(4a \cos \theta - l)$$

$$AD \cos \theta = 2a \cos \theta \quad M1$$

$$\text{Dividing: } 4a \tan \theta = \frac{4a \cos \theta - l}{\sin \theta}$$

$$l = \frac{4a \cos 2\theta}{\cos \theta} \quad A1$$

$$l = 4a \cos \theta - 4a \sin \theta \tan \theta$$

$$l = \frac{4a \cos 2\theta}{\cos \theta}$$

A1

5

or

Resolving perpendicular to S:

$$R \cos \theta = mg \cos 2\theta$$

(M1A1)

Moments about A:

$$R 2a \cos \theta = mg \frac{1}{2} l \cos \theta$$

(M1A1)

$$4a R = mgl$$

$$4a mg \cos 2\theta = mgl \cos \theta$$

$$l = \frac{4a \cos 2\theta}{\cos \theta}$$

(A1)

(5)

or

Resolving horizontally:

$$R \sin \theta = S \cos 2\theta$$

Both attempted for M1

Both correct for A1

(M1A1)

Resolving vertically:

$$R \cos \theta + S \sin 2\theta = mg$$

Moments about A:

$$R 2a \cos \theta = mg \frac{1}{2} l \cos \theta$$

(M1A1)

$$4a R = mgl$$

$$R \cos \theta + R \frac{\sin \theta}{\cos 2\theta} \sin 2\theta = 4a \frac{R}{l}$$

$$l = \frac{4a \cos 2\theta}{\cos \theta}$$

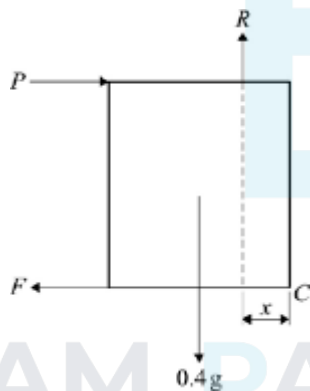
(A1)

(5)

[8]

Q2.

On point of sliding:



$$P = F$$

$$R = 0.4g$$

$$F = \mu R$$

$$P = 0.4g \mu$$

Obtaining P in terms of μ

B1

Moments about C:

$$P(0.28) + Rx = 0.4g(0.10)$$

M1 Forming a moment equation at least one term correct

M1

$$(0.4g \mu)(0.28) + (0.4g)x = 0.4g(0.10)$$

A1 – two terms correct

A2 – all terms correct x, μ only

A2,1

$$x > 0 \Rightarrow x = 0.1 - 0.28\mu > 0$$

$$\mu < \frac{0.1}{0.28} = \frac{10}{28} = \frac{5}{14}$$

Using $x > 0$

M1

$$k = \frac{5}{14} \text{ or } 0.357$$

CAO

A1

Alternative

$$\begin{array}{l} F = \mu R \\ \text{To slide: } R = 0.4g \\ F = p \end{array} \left. \vphantom{\begin{array}{l} F = \mu R \\ R = 0.4g \\ F = p \end{array}} \right\} P = 3.92\mu$$

Obtaining P in terms of μ

(B1)

Consider toppling: P needed

Take moments about C to find P :

$$0.4g(0.10) = P(0.28)$$

M1 – One side correct, A1 all correct

(M1A1)

$$1.4 = P$$

A1 – Correct P value

(A1)

$$P_{\text{sliding}} < P_{\text{toppling}}$$

$$\text{Slides before topples} \Rightarrow 3.92\mu < 1.4$$

Inequality for μ using $P_{\text{sliding}} < P_{\text{toppling}}$

(M1)

$$\mu < \frac{1.4}{3.92} = \frac{5}{14} = k$$



Accept 0.357 (NB M0 if inequality reversed in equation)

(A1)

[6]

Q3.

(a) $4\mathbf{i} + 4\mathbf{k}$

B1

1

(b)

$$\begin{vmatrix} \mathbf{i} & 1 & 2 \\ \mathbf{j} & 0 & 3 \\ \mathbf{k} & 0 & 0 \end{vmatrix} = 3\mathbf{k}$$

At least one $r \times F$ (or $F \times r$) correct

M1

$$\begin{vmatrix} \mathbf{i} & 1 & 2 \\ \mathbf{j} & 1 & -3 \\ \mathbf{k} & 0 & 0 \end{vmatrix} = -5\mathbf{k}$$

$$\begin{vmatrix} \mathbf{i} & 0 & 0 \\ \mathbf{j} & 0 & -1 \\ \mathbf{k} & 1 & 2 \end{vmatrix} = \mathbf{i}$$

$$\begin{vmatrix} \mathbf{i} & 0 & 0 \\ \mathbf{j} & 1 & 1 \\ \mathbf{k} & 1 & 2 \end{vmatrix} = \mathbf{i}$$

A1 three correct – all four correct for A2

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A2,1

Total = $2\mathbf{i} - 2\mathbf{k}$

Totalling their four determinants –

m1

dependent on first M1(max 3 / 5 for $F \times r$)

A1

5

(c) $\mathbf{P} = -4\mathbf{i} - 4\mathbf{k}$

ft part (a): $-1 \times$ their answer from (a)

B1F

Moment = $-2\mathbf{i} + 2\mathbf{k}$

B1F



ft part (b): $-1 \times$ their answer from (b)

Moment can be implied by appropriate sum equal to 0

Coords = (0, y, 0)

Hence $\begin{vmatrix} \mathbf{i} & 0 & -4 \\ \mathbf{j} & y & 0 \\ \mathbf{k} & 0 & -4 \end{vmatrix} = -2\mathbf{i} + 2\mathbf{k}$

Forming an $\mathbf{r} \times \mathbf{F}$ equation using their $\mathbf{P}$, point on y-axis and their moment

M1

Fully correct equation

A1

Determinant = $-4y\mathbf{i} + 4y\mathbf{k}$

Their $\mathbf{r} \times \mathbf{F}$ evaluated but j component must be 0

A1F

$$\therefore y = \frac{1}{2}$$

Coords at $(0, \frac{1}{2}, 0)$
CSO

A1

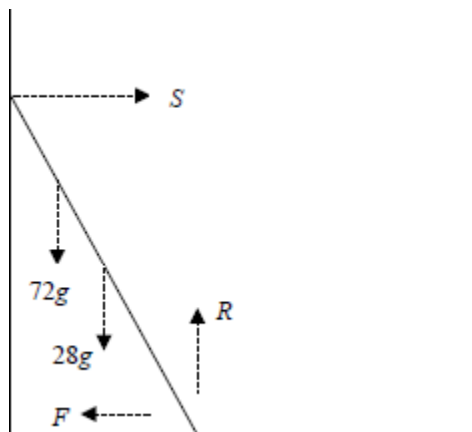
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[12]

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Q4.

(a)



force diagram

accept 'weight of man' or w_m etc for $72g$

B1 for any error

B2

- (b) (i) moments about P:
 $72g \times 6 \times \cos 69 + 28g \times 4 \times \cos 69$
3 terms including distance and angles

M1

$$= S \times 8 \times \sin 69$$

$$(432g + 112g)\cos 69 = 8 S \sin 69$$

A1 2 correct terms

A1A1

$$S = 255.8$$

accept division seen eg $\frac{544g}{8 \tan 69}$

$$= 256N$$

A1

4

- (ii) resolve vertically:
 $R = 28g + 72g$
 $= 100g$

B1

resolve horizontally:

$$S = F$$

B1

using $F = \mu R$:
 $\mu = 256 \div 100g$

M1

$$= 0.261$$

A1

4

[10]

Q5.

(a) $M = \left(\frac{4-2}{2}, \frac{-1+1}{2}, \frac{4+6}{2} \right) = (1, 0, 5)$
mid-point found

B1

$$\overrightarrow{PM} = - \frac{1}{AG} \begin{pmatrix} -2 \\ -1 \\ 4 \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \\ 5 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix}$$

B1

2

Alternative

$$\overrightarrow{PQ} = \begin{pmatrix} 4 \\ 1 \\ 6 \end{pmatrix} - \begin{pmatrix} -2 \\ -1 \\ 4 \end{pmatrix} = \begin{pmatrix} 6 \\ 2 \\ 2 \end{pmatrix}$$

(B1)

$$\overrightarrow{PM} = \frac{1}{2} \begin{pmatrix} 6 \\ 2 \\ 2 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix}$$

AG

(B1)

2

(b) Moment = $\mathbf{r} \times \mathbf{F}$

$$= \begin{vmatrix} \mathbf{i} & 3 & a \\ \mathbf{j} & 1 & 1 \\ \mathbf{k} & 1 & -2 \end{vmatrix}$$

attempt at $\mathbf{r} \times \mathbf{F}$ or $\mathbf{F} \times \mathbf{r}$

M1

$$= \begin{pmatrix} -3 \\ a+6 \\ 3-a \end{pmatrix}$$

one component correct $\Rightarrow$ A1

$\mathbf{F} \times \mathbf{r}$ attempt $\Rightarrow$ M1A1A0

A2,1

3

(c) Magnitude = $\sqrt{(-3)^2 + (a+6)^2 + (3-a)^2}$
attempt at magnitude of their moment

M1

$$\text{Hence } 9 + (a+6)^2 + (3-a)^2 = 50$$

forms equation $\text{magnitude}^2 = 50$

A1F

$$a^2 + 3a + 2 = 0$$

$$(a + 2)(a + 1) = 0$$

attempts to solve a quadratic – real roots

m1

$$a = -2 \text{ or } -1$$

both values obtained; CAO

No further penalty for $\mathbf{F} \times \mathbf{r}$ attempt which is correct

ie $(3, -a - 6, a - 3)$ as components

A1

4

[9]

Q6.

$$(a) \quad \begin{pmatrix} -2 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ -5 \end{pmatrix} + \begin{pmatrix} p \\ 1 \end{pmatrix} + \begin{pmatrix} -3 \\ -4 \end{pmatrix} = \begin{pmatrix} p-5 \\ -8 \end{pmatrix}$$

B1

1

- (b) (i) Parallel to y-axis $\Rightarrow p-5 = 0$
set $\mathbf{i}$ component = 0 (seen or implied)

M1

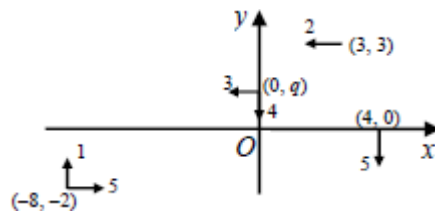
$$p = 5$$

A1

2

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(ii)



Moments about O for given system

$$-5(4) + 2(3) + 3q + 5(2) - 1(8)$$

$F \times d$ for at least four components

M1

$$= 3q - 12$$

A2,1F



–1 each type of error, ft (a), (b)(i)

(12 – 3q scores M1A2)

Moments about O for equivalent system

$$= -8(3)$$

$$= -24$$

±24 seen

ft (a) allow ± 3 × their **j** component

B1

$$\text{Hence } 3q - 12 = -24$$

attempt at moment equation – must see

clear use of Force × distance on RHS

M1

$$3q = -12$$

$$q = -4$$

ft error with p from (b)(i)

A1F

6

(c) $|C| = 24$

Should match part (b) – must be positive

B1F

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accept 'clockwise'

B1

2

[11]

Q7.

(a)
$$\begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 4 \\ 0 \\ -2 \end{pmatrix} + \begin{pmatrix} -3 \\ -1 \\ 2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

Clear use of $\sum F_i = 0$

B1

1

(b) $\mathbf{r}_1 \times \mathbf{F}_1 = \begin{vmatrix} \mathbf{i} & 0 & -1 \\ \mathbf{j} & 2 & 1 \\ \mathbf{k} & 1 & 0 \end{vmatrix} = \begin{pmatrix} -1 \\ -1 \\ 2 \end{pmatrix}$

Any attempt at $\mathbf{r} \times \mathbf{F}$ or $\mathbf{F} \times \mathbf{r}$

M1

$$\mathbf{r}_2 \times \mathbf{F}_2 = \begin{vmatrix} \mathbf{i} & 3 & 4 \\ \mathbf{j} & -1 & 0 \\ \mathbf{k} & 0 & -2 \end{vmatrix} = \begin{pmatrix} 2 \\ 6 \\ 4 \end{pmatrix}$$

$$\mathbf{r}_3 \times \mathbf{F}_3 = \begin{vmatrix} \mathbf{i} & 4 & -3 \\ \mathbf{j} & 0 & -1 \\ \mathbf{k} & -5 & 2 \end{vmatrix} = \begin{pmatrix} -5 \\ 7 \\ -4 \end{pmatrix}$$

-1 each 'type' of error

A3,2,1

$$\begin{pmatrix} -1 \\ -1 \\ 2 \end{pmatrix} + \begin{pmatrix} 2 \\ 6 \\ 4 \end{pmatrix} + \begin{pmatrix} -5 \\ 7 \\ -4 \end{pmatrix} = \begin{pmatrix} -4 \\ 12 \\ 2 \end{pmatrix}$$

Summing 3 vector product expressions

Follow through their three answers for $\mathbf{r} \times \mathbf{F}$

$\mathbf{F} \times \mathbf{r}$ loses final A1

m1
A1F

6

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(c) Resultant force = $\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$ and moment about O is non zero $\Rightarrow$ couple

E1 $\sum \mathbf{F} = 0$

E1 Moment $\neq 0$

E2,1

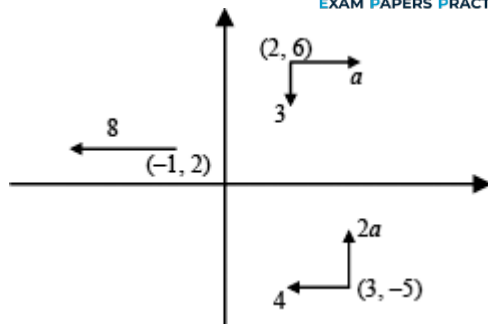
O is non zero $\Rightarrow$ couple

2

[9]

Q8.

(a)



Moments about O :

$$-3(2) - a(6) + 2a(3) - 4(5) + 8(2)$$

One correct $F \times d$ pairing seen
- 1 each type of error

M1
A2,1

$$= -6 - 6a + 6a - 20 + 16$$

$$= -10 \Rightarrow \text{no 'a'} \Rightarrow \text{independent}$$

'a' cancels / comment about no 'a'

A1

$$\text{Magnitude} = 10$$

Must be positive value stated;
ft single slip

A1F

5

Alternative:

$$\mathbf{r}_1 \times \mathbf{F}_1 = \begin{vmatrix} \mathbf{i} & -1 & -8 \\ \mathbf{j} & 2 & 0 \\ \mathbf{k} & 0 & 0 \end{vmatrix} = \begin{pmatrix} 0 \\ 0 \\ 16 \end{pmatrix}$$

One correct $\mathbf{r} \times \mathbf{F}$ seen

(M1)

$$\mathbf{r}_2 \times \mathbf{F}_2 = \begin{vmatrix} \mathbf{i} & 2 & a \\ \mathbf{j} & 6 & -3 \\ \mathbf{k} & 0 & 0 \end{vmatrix} = \begin{pmatrix} 0 \\ 0 \\ -6 - 6a \end{pmatrix}$$

Second $\mathbf{r} \times \mathbf{F}$ correct

(A1)

$$\mathbf{r}_3 \times \mathbf{F}_3 = \begin{vmatrix} \mathbf{i} & 3 & -4 \\ \mathbf{j} & -5 & 2a \\ \mathbf{k} & 0 & 0 \end{vmatrix} = \begin{pmatrix} 0 \\ 0 \\ 6a - 20 \end{pmatrix}$$

Total moment = $16 - 6 - 6a + 6a - 20$

$\Sigma(k \text{ components})$

(A1)

= -10 'a' cancels / no 'a'

Comment about 'a'

(A1)

Magnitude = 10

ft single slip

(A1F)

(5)

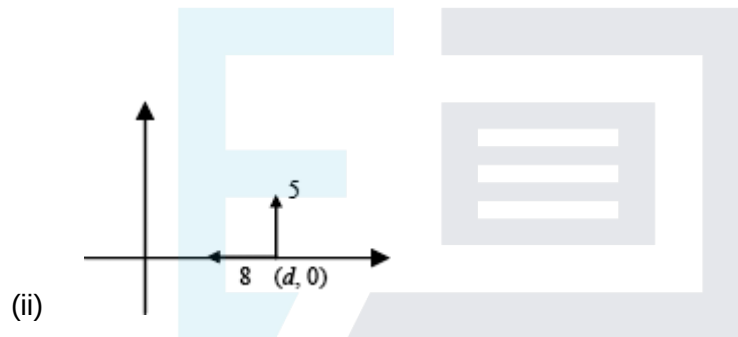
(b) (i) $\mathbf{F} = \begin{pmatrix} 4 \\ -3 \end{pmatrix} + \begin{pmatrix} -4 \\ 8 \end{pmatrix} + \begin{pmatrix} -8 \\ 0 \end{pmatrix} = \begin{pmatrix} -8 \\ 5 \end{pmatrix}$

Substitute $a = 4$ and add forces

M1

A1

2



Moments about O: $10 = 5d$

M1

y - component $\times d = \text{answer (a)}$

$d = -2$
ft (a)

A1F

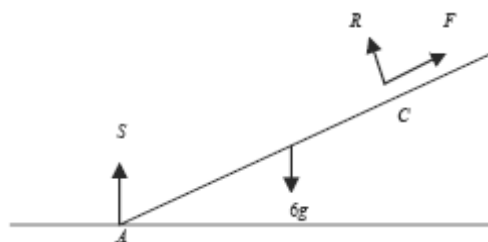
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[9]

Q9.

(a)



B1 for S and 6g (in correct place)

B1 for R and F or combined vertical force at C

B2

2

- (b) Moments about C:

$$3 \times S \times \cos 20 = 6g \times 1 \times \cos 20$$

M1 2 terms, 1 term correct

M1A1

$$S = 19.6 \text{ N or } 2g$$

A1

3

- (c) (i) Moments about A:

$$2 \times 6g \times \cos 20 = R \times 3$$

M1A1

$$R = 36.8 \text{ N}$$

$$\text{(or resolving, } R = 6g \cos 20 - S \cos 20 \\ = 4g \cos 20)$$

A1

- (ii) Resolve parallel to AB:

$$S \cos 70 + F = 6g \cos 70$$

M1

$$F = 4g \cos 70$$

$$= 13.4 \text{ N}$$

$$\text{(or } F = 6g \sin 20 - S \sin 20 = 4g \sin 20)$$

A1

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R, F not correct 0 marks in (c)(i) and (c)(ii)

Or

Moments about mid-point of rod:

$$2 \times S \times \cos 20 = P \times 1 \times \cos 20$$

$$P = 39.2 \text{ N or } 4g$$

(Or resolving vertically } P = 4g)

$$R = P \times \cos 20$$

$$= 36.8 \text{ N}$$

$$F = P \times \sin 20$$

$$= 13.4 \text{ N}$$

M1A1

A1

M1

A1

5

- (d) Using $F = \mu R$:

$$M1 \text{ '(c)(ii)' } = \mu \text{ '(c)(i)'}$$

M1

$$13.4 = \mu \times 36.8$$

$$\mu = 0.364 \text{ or } \tan 20$$

(condone $\geq$)

A1ft

2

[12]

Q10.

(a) Moments about C:

$$P \cos \theta \text{ seen}$$

B1

$$mga = P \cos \theta a$$

Forming moments equation – 2 terms

M1

$$P = \frac{mg}{\cos \theta}$$

A1

3

(b) Resolve $\leftrightarrow$

$$F = P \cos \theta \quad 1$$

Resolve in two directions

M1

Resolve $\uparrow$ $mg = R + P \sin \theta \quad 2$

Both equations correct

A1

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Friction law (sliding) $F = \mu R \quad 3$

Substituting 1 and 2 in 3:

$$P \cos \theta = \mu (mg - P \sin \theta)$$

Substituting in $F = \mu R$ – dep on first M1

m1

$$P \cos \theta + P \mu \sin \theta = \mu mg$$

$$P = \frac{\mu mg}{\cos \theta + \mu \sin \theta}$$

AG

A1

4

(c) Slides first $\Rightarrow$

$$\frac{\mu mg}{\cos \theta + \mu \sin \theta} < \frac{mg}{\cos \theta}$$

Set up inequality –
expression in (b) < expression in (a)(ii)

M1

$$\mu \cos \theta < \cos \theta + \mu \sin \theta$$

Correct simplification – remove fractions
ft parts (a) and (b)

A1F

$$\mu(\cos \theta - \sin \theta) < \cos \theta$$

$$\mu < \frac{\cos \theta}{\cos \theta - \sin \theta}$$

CAO ; Alternative: $\mu < \frac{1}{1 - \tan \theta}$

A1

3

- (d) Inequality independent of mass, so no change
No change (E1) and reason (E1)
ft error in (c); must give consistent reason
If no reason, E0

E2,1F

2

[12]

Q11.

(a) $\mathbf{r}_1 \times \mathbf{F}_1 = \begin{bmatrix} \mathbf{i} & 1 & 2 \\ \mathbf{j} & 0 & 0 \\ \mathbf{k} & 3 & a \end{bmatrix} = \begin{pmatrix} 0 \\ 6-a \\ 0 \end{pmatrix}$

Attempt at $\mathbf{r}_1 \times \mathbf{F}_1$ – one comp correct

M1

Fully correct

A1

$$\mathbf{r}_2 \times \mathbf{F}_2 = \begin{bmatrix} \mathbf{i} & -1 & -2 \\ \mathbf{j} & 2 & 1 \\ \mathbf{k} & 0 & 3 \end{bmatrix} = \begin{pmatrix} 6 \\ 3 \\ 3 \end{pmatrix}$$

Attempt at $\mathbf{r}_2 \times \mathbf{F}_2$ – one comp correct

M1A1

$$\begin{pmatrix} 0 \\ 6-a \\ 0 \end{pmatrix} + \begin{pmatrix} 6 \\ 3 \\ 3 \end{pmatrix} = \begin{pmatrix} 6 \\ 9-a \\ 3 \end{pmatrix}$$

Totalling $r_1 \times F_1 + r_2 \times F_2$; AG

A1

Note $F \times r$ scores M1A0M1A1A0

5

- (b) (i) Magnitude of couple = 7
- $\Rightarrow$

$$6^2 + (9 - a)^2 + 3^2 = 7^2$$

Forming equation – using part (a)

M1

$$(9 - a)^2 = 4$$

Solving – to obtain two values

M1

$$\therefore 9 - a = 2 \text{ or } 9 - a = -2$$

$$a = 7 \text{ or } a = 11$$

AG (7)

A1, A1

Special case (max 2):

$$\begin{pmatrix} 6 \\ 2 \\ 3 \end{pmatrix}$$

If $a = 7$, then resultant moment =

$$\text{Magnitude} = \sqrt{6^2 + 2^2 + 3^2} = 7$$

(M1)(A1)

4

$$a = 7 \Rightarrow \mathbf{F}_1 + \mathbf{F}_2 = \begin{pmatrix} 2 \\ 0 \\ 7 \end{pmatrix} + \begin{pmatrix} -2 \\ 1 \\ 3 \end{pmatrix}$$

(ii)

Attempt at $\mathbf{F}_1 + \mathbf{F}_2$

M1

$$= \begin{pmatrix} 0 \\ 1 \\ 10 \end{pmatrix}$$

 $\mathbf{F}_1 + \mathbf{F}_2$ correct

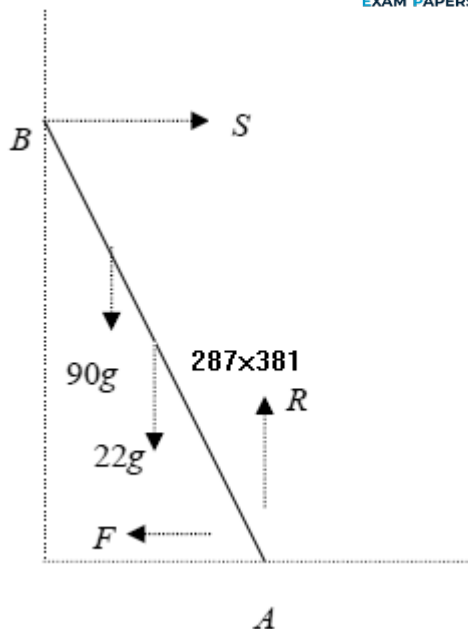
A1

2

[11]

Q12.

(a)



Resolve vertically:

$$R = 22g + 90g$$

$$= 112g$$

B1

Using $F = \mu R$:

M1

$$F = 0.6R$$

$$F = 0.6 \times 112g$$

*Needs $0.6 \times 112g$ or 0.6×1097.6
NOT 0.6×1097 unless 658.56 seen*

A1

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$$= 67.2g \text{ or } 658.56$$

$$F = 659 \text{ N}$$

AG (659 must be shown from correct working)

A1

4

(b) Resolve horizontally: $S = F$

B1

Moments about A:

$$90g \times 5 \times \cos \theta + 22g \times 3 \times \cos \theta$$

M1 (one term, force $\times$ distance $\times$ cos or sin)

M1A1

$$= 67.2g \times 6 \times \sin \theta$$

A1

$$450g + 66g = 403.2g \tan \theta$$

$$\tan \theta = \frac{516}{403.2}$$

$$\theta = 52.0^\circ$$

accept 52

Alternative: or moments about B:

M1 A2, 1 or 0 for four-term moment equation

+ M1 for rearranging etc (dep on 4 term)

+ A1 for answer

A1

5

[9]

Q13.

$$(a) \quad \mathbf{F} = \begin{pmatrix} 1 \\ 5 \\ -3 \end{pmatrix} + \begin{pmatrix} 2 \\ -7 \\ -1 \end{pmatrix} + \begin{pmatrix} 0 \\ 4 \\ -2 \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \\ -6 \end{pmatrix}$$

Correct total

B1

$$|\mathbf{F}| = \sqrt{3^2 + 2^2 + 6^2}$$

Attempt to find $|\mathbf{F}|$

M1

$$= 7$$

AG

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A1

3

$$(b) \quad (i) \quad \mathbf{r} = \begin{pmatrix} 6 \\ 4 \\ -7 \end{pmatrix} - \begin{pmatrix} 3 \\ 6 \\ -3 \end{pmatrix} = \begin{pmatrix} 3 \\ -2 \\ -4 \end{pmatrix}$$

Correct $\mathbf{r}$

B1

$$\text{Moment} = \mathbf{r} \times \mathbf{F}$$

$$= \begin{vmatrix} \mathbf{i} & 3 & 0 \\ \mathbf{j} & -2 & 4 \\ \mathbf{k} & -4 & -2 \end{vmatrix}$$

Attempt at $\mathbf{r} \times \mathbf{F}$ or $\mathbf{F} \times \mathbf{r}$

M1

$$= \begin{pmatrix} 20 \\ 6 \\ 12 \end{pmatrix}$$

One component correct $\Rightarrow$ A1

All components correct $\Rightarrow$ A2

A2, 1, 0

4

SC1: $\mathbf{F} \times \mathbf{r} \Rightarrow$ M1A1A0

SC2: Use of $\begin{pmatrix} -3 \\ 2 \\ 4 \end{pmatrix}$ to get $\begin{pmatrix} -20 \\ -6 \\ -12 \end{pmatrix}$ scores
B0 M1 A1 A1F

SC3: Use of $\mathbf{r} = \begin{pmatrix} 3 \\ 6 \\ -3 \end{pmatrix}$ to get $\begin{pmatrix} 0 \\ 6 \\ 12 \end{pmatrix}$ scores
B0 M1 A1F A0

(ii) Moment = $\begin{pmatrix} 20 \\ 6 \\ 12 \end{pmatrix}$
ft (b) (i)

B1F

since resultant of other two forces acts through given point (therefore 0 moment)

E1

2

[9]

Q14.

(a) Let resultant be $\begin{pmatrix} X \\ Y \end{pmatrix} = R$

$$X = 8 + 6 - 15 \cos \theta$$

$$Y = 1 + 2 - 15 \sin \theta$$

Attempt at X and Y; must involve use of $15 \sin \theta$ or $15 \cos \theta$

M1

with $\cos \theta = \frac{8}{10}$ and $\sin \theta = \frac{6}{10}$

or $\theta = 36.9^\circ$

Either 12 or 9 seen as components of the 15 N force

A1

$\Rightarrow X = 2, Y = -6$

Both X and Y correctly evaluated including direction

A1

$|R| = \sqrt{2^2 + 6^2} = \sqrt{40}$

Attempt at |R|

$= 2\sqrt{10}$

AG; must see $\sqrt{40}$ or $\sqrt{4 \times 10}$

m1

A1

Alternative – using diagrams:



4 components shown

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12 or 9 seen

(A1)

Resultant components - correct direction shown

(A1)

As above

(m1)

As above

(A1)

5

(b) (i) Moments about O for system

Attempt at moments for system

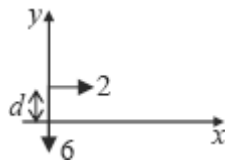
M1

$$20 + 2(8) - 8(6) = -12$$

-1 each error or omission

A2, 1, 0

(ie 12 Nm clockwise)



Moment of resultant

$$2d = 12$$

Form equation – must be of form
 $x\text{-component} \times d = \text{moment for system}$

M1

$$d = 6$$

A1

5

$$(ii) \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 6 \end{pmatrix} + t \begin{pmatrix} 1 \\ -3 \end{pmatrix}$$

Correct structure on RHS ($a + tb$)

M1

$$\begin{pmatrix} 0 \\ 6 \end{pmatrix}; \text{ft } d \text{ value from (b)(i)}$$

A1F

$$\begin{pmatrix} 1 \\ -3 \end{pmatrix} \text{ OE; ft components from (a)}$$

Condone omission of $\begin{pmatrix} x \\ y \end{pmatrix}$ or r on LHS

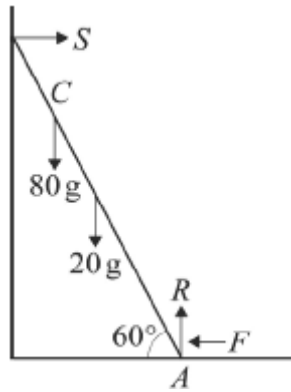
A1F

3

[13]

Q15.

(a)



B1 for any 4 correct

B2

2

(b) Resolve vertically:

$$R = 20g + 80g$$

Must see $20g + 80g$ or $100g$ to obtain any marks in (b)

$$= 100g$$

B1

Using $F = \mu R$:

Dep on B1

m1

$$\begin{aligned} F &= 0.4 \times 100g \\ &= 40g \text{ or } 392 \text{ N} \end{aligned}$$

AG

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A1

3

(c) Resolve horizontally:

$$S = 40g$$

B1

Moments about A:

$$80gx \cos 60 + 20g \cdot 2 \cos 60$$

M1 for 3 terms, all moments

M1A1

$$= S \cdot 4 \cos 30$$

A1

$$40gx + 20g = 138.56g$$

$$x = \frac{118.56}{40}$$

Dep on M1

m1

$$= 2.96 \text{ m}$$

Accept $2\sqrt{3} - \frac{1}{2}$

A1

6

[11]

Q16.

(a) Couple $\Rightarrow \sum \text{horizontal component} = 0$

$$\sum \text{vertical component} = 0$$

Vertically:

$$2\sqrt{3} \cos 60^\circ - Q \cos 30^\circ = 0$$

$$\sum \text{vertical component} = 0$$

M1

$$\therefore Q = 2$$

AG

A1

Horizontally:

$$P - 2\sqrt{3} \sin 60^\circ - Q \sin 30^\circ = 0$$

$$\sum \text{horizontal component} = 0$$

M1

one component correct (condone $\pm$)

A1

$$\therefore P = 4$$

A1

5

(b) Moments about B:

(N.B clockwise – ve/ anticlockwise + ve
in solution below)

$$2\sqrt{3} \sin 60^\circ (4) - 4 (5)$$

Evidence of force $\times$ perp distance

One term correct; ft error with P

M1A1ft

$$= -8$$

Magnitude = 8

A1ft

3

Or

Moments about A:

$$-2\sqrt{3} \sin 60^\circ (1) - 2 \sin 30^\circ (5)$$

Evidence of force $\times$ perp distance
One term correct

(M1A1)

$$= -8$$

Magnitude = 8

No ft for Q

(A1)

Or

Moments about C:

$$-4 (1) - 2 \sin 30^\circ (4)$$

Evidence of force $\times$ perp distance
One term correct, ft error with P

(M1A1ft)

$$= -8$$

Magnitude = 8

(A1ft)

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Moments about centre of rod

$$-P(2.5) - Q(2.5 \sin 30^\circ) + 2\sqrt{3} (1.5 \sin 60^\circ)$$

Evidence of force $\times$ perp distance
One term correct, ft error with P

M1A1ft

$$= -8$$

Magnitude = 8

A1ft

Or

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} \times \begin{bmatrix} 4 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ -1 \end{bmatrix} \times \begin{bmatrix} -3 \\ \sqrt{3} \end{bmatrix} + \begin{bmatrix} 0 \\ -5 \end{bmatrix} \times \begin{bmatrix} -1 \\ -\sqrt{3} \end{bmatrix}$$

Evidence of $\mathbf{r} \times \mathbf{F}$

M1

$$= (0 - 3 - 5)\mathbf{k}$$

one value correct

A1ft

$$= -8\mathbf{k} \text{ Magnitude} = 8$$

ft P value

A1ft

SC Max M1A0A0 for candidates who form an equation in part (b) without using a variable for couple
i.e. $4(2.5) + 2\sqrt{3}(1.5\sin 60^\circ) = 2(2.5\sin 30^\circ)$

- (c) Clockwise
ft answer (b) if directions all clear

B1ft

1

[9]

Q17.

(a) $\overrightarrow{AB} = \begin{pmatrix} 2 \\ 3 \\ -6 \end{pmatrix}$

B1

1

(b) $\overrightarrow{AB} \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & 2 & 2 \\ \mathbf{j} & 3 & -1 \\ \mathbf{k} & -6 & 4 \end{vmatrix}$

Attempt at $\mathbf{r} \times \mathbf{F}$ or $\mathbf{F} \times \mathbf{r}$
M0 if no evidence of $\mathbf{i}, \mathbf{j}, \mathbf{k}$ components

M1

$$= \begin{pmatrix} 6 \\ -20 \\ -8 \end{pmatrix}$$

One component correct = A1
Follow through $\overrightarrow{AB}$
[If $\mathbf{F} \times \mathbf{r}$ M1, A1, A0] max

A2,1,0ft

3

(c) $\sqrt{6^2 + 20^2 + 8^2} = \sqrt{500}$

M1

$$= 10\sqrt{5} \text{ N}$$

AG must see $\sqrt{500}$ to award A1

A1

2

(d)

$$\sin \theta = \frac{10\sqrt{5}}{\begin{vmatrix} 2 \\ 3 \\ -6 \end{vmatrix} \begin{vmatrix} 2 \\ -1 \\ 4 \end{vmatrix}}$$

Use of $\sin \theta = \frac{\mathbf{a} \times \mathbf{b}}{|\mathbf{a}||\mathbf{b}|}$ with correct vector pair

M1

$\sqrt{49}, 7$ or $\sqrt{21}$ seen

B1

$$= \frac{10\sqrt{5}}{7\sqrt{21}}$$

Correct values ft their $\vec{AB}$

A1ft

$$\theta = 44^\circ$$

ft their $\vec{AB}$

A1ft

SC if $90^\circ - \theta$ found (wrong angle – correct triangle)
ie 46° then award M1B1A1A0 Max

Alternative

$$\vec{AB} \cdot \mathbf{F} = \begin{pmatrix} 2 \\ 3 \\ -6 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ -1 \\ 4 \end{pmatrix} = -23$$

Their $\vec{AB} \cdot \mathbf{F}$

B1ft

$$\cos \theta = \frac{-23}{\begin{vmatrix} 2 \\ 3 \\ -6 \end{vmatrix} \begin{vmatrix} 2 \\ -1 \\ 4 \end{vmatrix}} = \frac{-23}{7\sqrt{21}}$$

$\theta = \frac{|a \cdot b|}{|a||b|}$
 use of cos with correct vector pair
 ft their $\overrightarrow{AB}$.
 (May not be explicitly seen)

M1A1ft

$$\theta = \cos^{-1}\left(\frac{-23}{7\sqrt{21}}\right) = 135.8^\circ \dots$$

$$\therefore \text{Required angle} = 180^\circ - 135.8^\circ = 44^\circ$$

ft their $\overrightarrow{AB}$

A1ft

N.B Use of $\sin\theta$ / $\cos\theta$ must be consistent with method chosen for M1

4

[10]

Q18.

(a) $M(AB)$
 $(0.8 \times 1.0) \rho \times 0.5 + (2.0 \times 0.2) \rho \times 2.0 = 1.2 \rho \times \bar{x}$

M1 all terms
 -1 EE from A2

M1A2

$$\bar{x} = 1, M \text{ on } CH$$

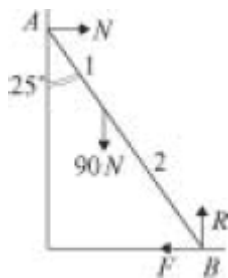
reference to position of M on CH needed

A1

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4

(b)



(i) $M(B) : 90 \times 2 \times \sin 25^\circ = N \times 3 \times \cos 25^\circ$
 Must attempt 'components' for M1

M1A1

$$N = 60 \tan 25^\circ$$

A1

(ii) $F = N$ $R = W = 90$

B1

$$F \leq \mu R$$

$$60 \tan 25^\circ \leq \mu \times 90$$

Condone equality for M1, correct

Substitution for A1

M1A1

Correct use of inequality throughout

A1

$$u \geq \frac{2}{3} \tan 25^\circ$$

4

[11]

Q19.

(a) (i)
$$\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \begin{pmatrix} 4 \\ -3 \\ 5 \end{pmatrix} + \mathbf{F} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

sum of forces = 0 must be seen for M1

M1

$$\Rightarrow \mathbf{F} = \begin{pmatrix} -5 \\ 1 \\ -8 \end{pmatrix}$$

± (5i - j + 8k) seen

B1

correct sign

A1

3

(ii) $|\mathbf{F}| = \sqrt{5^2 + 1^2 + 8^2} = \sqrt{90} = 3\sqrt{10}$

√(their F components)

M1

AG

A1

2

(b) Moment = $\mathbf{r} \times \mathbf{F}$

$$= \begin{vmatrix} \mathbf{i} & 1 & 1 \\ \mathbf{j} & -1 & 2 \\ \mathbf{k} & 6 & 3 \end{vmatrix} + \begin{vmatrix} \mathbf{i} & 0 & 4 \\ \mathbf{j} & 3 & -3 \\ \mathbf{k} & -2 & 5 \end{vmatrix} + \begin{vmatrix} \mathbf{i} & 0 & -5 \\ \mathbf{j} & 3 & 1 \\ \mathbf{k} & -2 & -8 \end{vmatrix}$$

attempt at one $\mathbf{r} \times \mathbf{F}$

M1

(all attempted)

M1

$$= \begin{pmatrix} -15 \\ 3 \\ 3 \end{pmatrix} + \begin{pmatrix} 9 \\ -8 \\ -12 \end{pmatrix} + \begin{pmatrix} -22 \\ 10 \\ 15 \end{pmatrix}$$

any three components correct

A1ft

all components correct

A1ft

$$= \begin{pmatrix} -28 \\ 5 \\ 6 \end{pmatrix}$$

sum of vectors; ft their $\mathbf{F}$ from part (a)

A1ft

5

1st Alternative for (b):

$$\overline{QP} = \begin{pmatrix} 1 \\ -4 \\ 8 \end{pmatrix}$$

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intention to use $\mathbf{r} \times \mathbf{F}$ about Q

(M1)

$\overline{QP}$ obtained correctly

(A1)

Moments about Q

$$QP \times \mathbf{F}_1 = \begin{vmatrix} \mathbf{i} & 1 & 1 \\ \mathbf{j} & -4 & 2 \\ \mathbf{k} & 8 & 3 \end{vmatrix} = \begin{pmatrix} -28 \\ 5 \\ 6 \end{pmatrix}$$

determinant attempted

(M1)

one component correct

all correct

(A1)

(A1)

(5)

2nd Alternative for (b):

$$\overrightarrow{PQ} = \begin{pmatrix} -1 \\ 4 \\ -8 \end{pmatrix}$$

intention to use $\mathbf{r} \times \mathbf{F}$ about P

(M1)

$\overrightarrow{PQ}$ *obtained correctly*

(A1)

$$\begin{vmatrix} \mathbf{i} & -1 & -5 \\ \mathbf{j} & 4 & 1 \\ \mathbf{k} & -8 & -8 \end{vmatrix} = \begin{pmatrix} -24 \\ 32 \\ 19 \end{pmatrix}$$

one determinant correct

(M1)

$$\begin{vmatrix} \mathbf{i} & -1 & 4 \\ \mathbf{j} & 4 & -3 \\ \mathbf{k} & -8 & 5 \end{vmatrix} = \begin{pmatrix} -4 \\ -27 \\ -13 \end{pmatrix}$$

both correct

(A1)

$$\begin{pmatrix} -24 \\ 32 \\ 19 \end{pmatrix} + \begin{pmatrix} -4 \\ -27 \\ -13 \end{pmatrix} = \begin{pmatrix} -28 \\ 5 \\ 6 \end{pmatrix}$$

all correct

(A1)

(5)

[10]

Q20.

- (a) On point of toppling, take moments about bottom right corner

attempt at moments

M1

$$W(2a) = P \cos \theta (8a)$$

A1 each side

A1,A1

$$P = \frac{W}{4 \cos \theta}$$

A1

4

- (b) On point of sliding vertically, $N + P \sin \theta = W$

M1A1

horizontally, $F = P \cos \theta$

M1A1

friction $F = \mu N$

$$\Rightarrow P \cos \theta = \mu(W - P \sin \theta)$$

substitute; use of $F = \mu N$

M1A1

$$P \cos \theta = \mu W - \mu P \sin \theta$$

$$P(\cos \theta + \mu \sin \theta) = \mu W$$

$$P = \frac{\mu W}{\cos \theta + \mu \sin \theta}$$

AG

A1

7

- (c) Slides before topples $\Rightarrow$

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$$\frac{\mu W}{\cos \theta + \mu \sin \theta} < \frac{W}{4 \cos \theta}$$

inequality formed

M1

$$4\mu \cos \theta < \cos \theta + \mu \sin \theta$$

elimination of fractions / cancel W

A1

$$4\mu < 1 + \mu \tan \theta$$

$\div$ by $\cos \theta$ and use of $\tan \theta = 1$

A1

$$\tan \theta = 1 \Rightarrow 3\mu < 1$$

collect μ terms

M1

$$\mu < \frac{1}{3}$$

A1

5

[16]

Q21.

(a)

$$\begin{array}{rrr}
 \mathbf{i} & \mathbf{j} & \mathbf{k} \\
 2 & & \\
 3 & -5 & a \\
 b & +5 & -2 \\
 \hline
 0 & 0 & 0
 \end{array}$$

$$\sum \text{forces} = 0$$

$$\Rightarrow b = -5, a = 2$$

Both correct

M1

A1

2

(b)

$$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 1 & 0 \\ 2 & 0 & 0 \end{vmatrix} + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -1 & 2 & 1 \\ -5 & 5 & -2 \end{vmatrix}$$

Intention to find $\mathbf{r} \times \mathbf{F}$ [or $\mathbf{F} \times \mathbf{r}$]

M1

Two correct non zero determinants attempted

A1

$$= -2\mathbf{k} - 9\mathbf{i} - 7\mathbf{j} + 5\mathbf{k}$$

At least two non zero terms correct

A1

$$= -9\mathbf{i} - 7\mathbf{j} + 3\mathbf{k}$$

Fully simplified. Lose 1 if $\mathbf{F} \times \mathbf{r}$ found

A1

4

[6]

Q22.

(a) (i)
$$\begin{pmatrix} 8+6+(-2)+0 \\ 4+5-2-2 \end{pmatrix} = \begin{pmatrix} 12 \\ 5 \end{pmatrix}$$

Adding forces

$R_x = 12 \quad R_y = 5$

M1 A1

$|F| = \sqrt{12^2 + 5^2} = 13$

Follow through mis-totals – use of Pythagoras' rule

A1ft

3

(ii) Moment of forces about O

$= -6(3) - 2(3) - 2(4) + 2(4)$

One term correct

M1

All terms correct [NB $\pm$ can be reversed]

A1

$-18 - 6 - 8 + 8$

-24

$12d = -24$

Equation formed $R_x d = \text{Total moment}$

M1

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$\therefore d = 2$

Printed answer

A1

4

Alternative

Total Moment = -24

(M1) (A1)

Equation of line of action is
$$y = \frac{5}{12} \left(x + \frac{24}{5} \right)$$

(M1)

When $x = 0$, $y = 2$

(A1)

(b) $|c| = 24$

Follow through (a)(ii)
ie |their total moment|

B1ft



Accept 'clockwise'
Follow through their direction from (a)(ii)

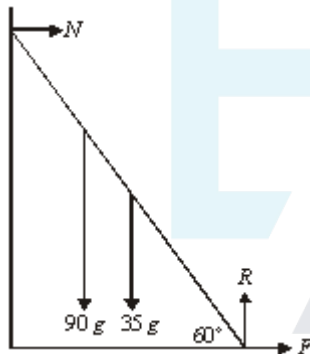
B1ft

2

[9]

Q23.

(a)



All forces shown and in correct direction (no extras)

B1

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1

(b) $R = 125 \text{ g}$ ($= 1225$)

B1

$$F = 0.3 \times R$$

Condone inequality

M1

$$F = 367.5 \text{ N}$$

ft slip, both vertical forces present
(g missing B0 M1 A1F)

A1F

3

(c) M (ground)

$$35 \text{ g} \times 1.5 \cos 60^\circ + 90 \text{ g} \times x \times \cos 60^\circ \\ = N \times 3 \cos 30^\circ$$

*M1 attempt at moments eqn.,
accept one force missing.
-1 each term missing or incorrect.
Condone repeated error, g missing or sin/cos mix.*

M1A2

$$F = N$$

B1

Substitute to find x

Subs. of candidate's N

m1

$$x = 1.582 \text{ metres}$$

Accept 1.6

A1

6

[10]

Q24.

(a) (i) Distance = $4 + \frac{1}{3} \times 6 = 6$

M1A1

2

(ii)

Shape	Mass	Dist from AB	Mass × Dist
ABEC'	24ρ	2	48ρ
C'ED	36ρ	6	216ρ
ABED	60ρ	$\bar{X}$	$60\rho \bar{X}$
	$\therefore 60\rho \bar{X}$	$= 264\rho$	
	$\bar{X}$	$= 4.4$	

no penalty if ρ omitted

M1

CAO; AG

A1

2

(iii)

Shape	Mass	Dist from AD	Mass × Dist
ABEC'	24ρ	3	72ρ
C'ED	36ρ	2	72ρ
ABED	60ρ	$\bar{Y}$	$60\rho \bar{Y}$
	$\therefore 60\rho \bar{Y}$	$= 144\rho$	

$$\bar{Y} = 2.4$$

A1
2

- (b) Moments about B : $K = 0.044 \times 2 = 0.088$
 $M1A0$ for 8.8

M1A1
2

[8]

Q25.

- (a) $X\mathbf{i} + Y\mathbf{j} = 4\mathbf{i} + 5\mathbf{j} + 2\mathbf{i} - \mathbf{j} - 3\mathbf{i} + 2\mathbf{j}$
 $= 3\mathbf{i} + 6\mathbf{j}$

A1
1

- (b) (i) Moments about O

$$= 5 \times 1 - 4 \times 2 + 2 \times 1 - 1 \times 3 + 3 \times 1 - 2 \times 2$$

-1 each error

M1

$$= -5$$

$$\text{magnitude} = 5$$

A2,1,0

A1
4

- (ii) Clockwise
ft consistent with (b)(i)

A1F
1

- (c) $3d = 5$
May assume clockwise + ve.

M1A1F

$$d = \frac{5}{3} \quad (1.67)$$

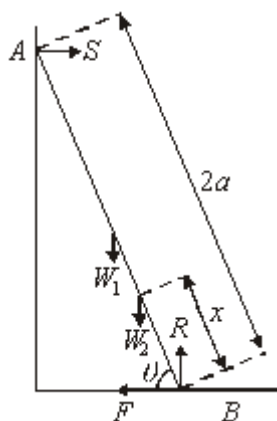
Must be consistent ft on (b)

A1F
3

[9]

Q26.

(a)



–1 each error. F and R may be combined as a single reaction force for full credit.

–1 for vertical force shown at A unless explained that this equals zero

A2,1,0

2

(b) Moments about B

$$S \cdot 2a \sin \theta = W_1 a \cos \theta + W_2 x \cos \theta$$

M1A1

$$S = \frac{1}{6} \left(W_1 + W_2 \frac{x}{a} \right)$$

(use of $\tan \theta = 3$)

m1A1

4

(c) $R = W_1 + W_2$

B1

$$F = S$$

B1

$$F \leq \mu R$$

$$\frac{1}{6} \left(W_1 + W_2 \frac{x}{a} \right) \leq (W_1 + W_2)$$

M1

$$x \leq \frac{a(4W_1 + 9W_2)}{5W_2}$$

CAO; AG

A1

4

- (d) For the ladder to remain in equilibrium with the man at the top

$$2a \leq \frac{a(4W_1 + 9W_2)}{5W_2}$$

M1

$$10W \leq 4W + 9W$$

$$W \leq 4W$$

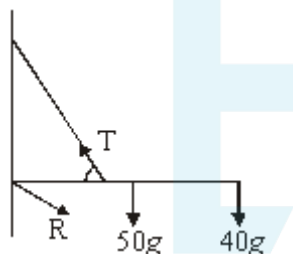
A1

3

[13]

Q27.

(a)



-1 for each error
 T, R acceptable as components

B2,1,0

2

- (b) Moments about A:

$$T \times 0.8 \sin \theta = 50g \times 1 + 40g \times 2$$

A2, 1, 0 (-1 each error)

M1A2

$$T = \frac{130g}{0.8 \sin \theta}$$

$$T < 2000$$

inequality used

M1

$$\frac{130g}{0.8 \sin \theta} < 2000$$

OE

M1

$$\theta > 52.8$$

*ft one error in T
dependent on **both** immediately preceding
M1 marks*

A1F

6

[8]

Q28.

(a)
$$\begin{pmatrix} 2 & a & -3 \\ -8 & b & -4 \end{pmatrix} = \begin{pmatrix} 3 \\ -6 \end{pmatrix}$$

M1

$$\begin{pmatrix} a-1 \\ b-12 \end{pmatrix} = \begin{pmatrix} 3 \\ -6 \end{pmatrix}$$

A1

$$\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 4 \\ 6 \end{pmatrix}$$

A1

(b) (i) Moments about O:

may assume clockwise positive must be consistent

M1

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$$6 \times 3 - 4 \times 2 - 3 \times 2 - 4 \times 4 = 12$$

at least 2 correct LHS terms

A1

correct RHS

A1

magnitude = 12

*must state magnitude for this mark
(even if taken "clockwise positive")*

A1

4

(ii) Clockwise

A0 if no M1 scored in (b)(i)

A1

(c) $-3d = -12$

M1

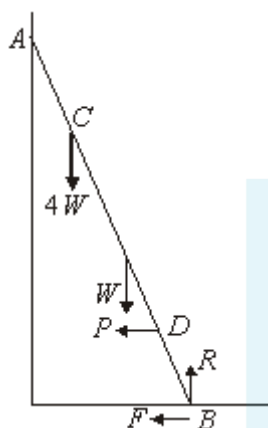
$d = 4$

A1

2

[10]

Q29.



(a) Resolving $\uparrow R = 5W$

B1

Ladder in limiting equilibrium

$F = \mu R$

M1

$$\begin{aligned} & \frac{11}{40} \times 5W \\ &= \frac{5W}{4} \end{aligned}$$

AG

A1

3

(b) Moments about A (or other appropriate point)

M1

$$\begin{aligned} 4W \cos \theta + W \times 2 \cos \theta + P \times 3 \sin \theta + F \times 4 \sin \theta \\ = R \times 4 \cos \theta \end{aligned}$$

(-1 per error)

A3,2,1

$$\Rightarrow 6W \cos\theta + 3P \sin\theta + \frac{5W}{4} \times 4 \sin\theta = 5W \times 4 \cos\theta$$

$$\Rightarrow 3P \times \frac{12}{13} + 5W \times \frac{12}{13} = 14W \times \frac{5}{13}$$

$$\text{use of } \sin\theta = \frac{12}{13} \text{ etc}$$

A1

$$\Rightarrow 36P + 60W = 70W$$

$$\Rightarrow 36P = 10W$$

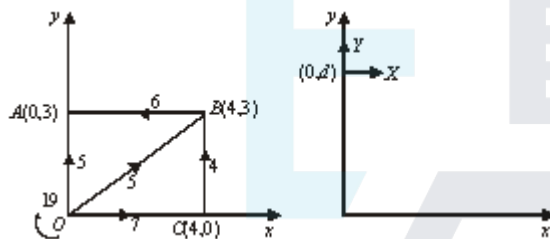
$$\Rightarrow P = \frac{5W}{18}$$

A1F

6

[9]

Q30.



(a) $X = 7 - 6 + 5\cos\theta$

Correct at this stage

M1A1

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$$= 1 + 5 \times \frac{4}{5}$$

$$= 5$$

$$Y = 4 + 5 + 5 \sin\theta$$

M1

$$= 9 + 5 \times \frac{3}{5}$$

$$= 12$$

(both X and Y correct)

A1

$$\therefore \text{Resultant} = \sqrt{5^2 + 12^2}$$

$$= 13$$

CAO

A1

(b) (i) $Xd = -4 \times 4 - 6 \times 3 + 19$
(for Xd)

M1A1

$5d = -15$
 $1^{st} 2 \text{ terms RHS}$

A1

$d = -3$
(+19)

A1

$\therefore$ line cuts axis at $(0, -3)$

CAO

[candidate may use "anticlockwise + ve" convention for full marks]

A1

5

(ii) Gradient line of action + $\frac{Y}{X} = \frac{12}{5}$

M1A1F

$\therefore y = \frac{12}{5}x - 3$
ft from (a)

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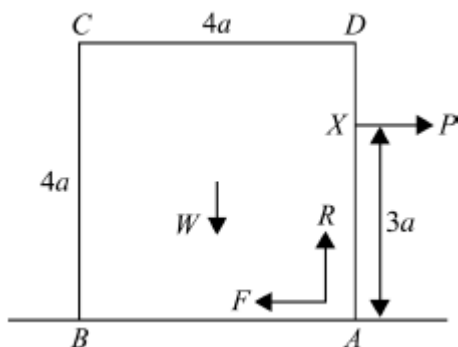
A1F

3

(or any acceptable equivalent e.g. $5y = 12x - 15$ etc)

[13]

Q31.



(a) Block sliding $\therefore P > F$

M1

$$R = W$$

B1

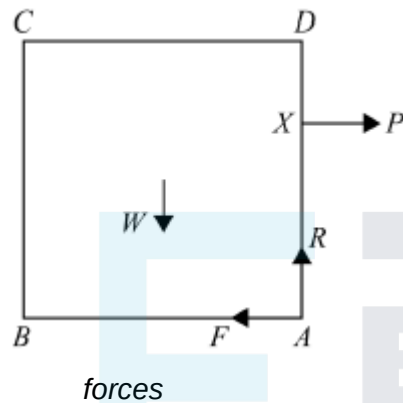
$$F = \mu R$$

$$\text{hence } P > \mu W$$

A1

3

(b) (i)



A1

1

(ii) $P \times 3a \geq W \times 2a$

M1

$$\Rightarrow P \geq \frac{2W}{3} \quad \text{allow = for A1}$$

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A1

2

(c) If $\mu = 0.6$, slides when $P > 0.6W$, if not toppled previously

B1

If $\mu = 0.667$, topples when $P > 0.667W$ if not started to slide

B1

$\therefore$ slides first

B1

3

[9]

Q32.

(a) $X = 2W$

B1

$$Y = W$$

B1

2

(b) Moments about A:

$$Wa \sin \theta = 2W \cdot 2a \cos \theta$$

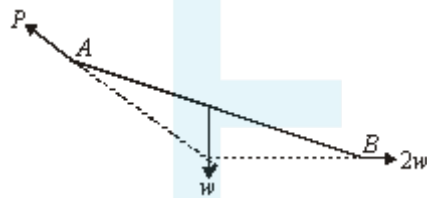
M1A1

$$\theta = \tan^{-1} 4 \ (\approx 76.0^\circ)$$

A1

3

(c) P must pass through the point of intersection of the lines of action of W and $2W$



Lines of action of W , $2W$ clearly indicated.

B1

Line of action of P clearly passing through the intersection of lines of action of W , $2W$.

B1

2

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[7]

Q33.

(a) $X = 4 + 3 + 3 + 2 = 12$

B1

$$Y = 3 + 4 + (-3) + 1 = 5$$

B1

$$F = \sqrt{(12^2 + 5^2)}$$

M1

$$= 13$$

Full credit if 5, 12, 13 seen

A1

- (b) Moments clockwise about O:

$$3 \times 2 + 3 \times 3 + 3 \times 4 - 1 \times 3 = 24$$

M1A1

$$- 5d = 24$$

M1

$$d = - 4.8$$

Or anticlockwise – must be consistent throughout

A1

4

- (c) L = 24 clockwise

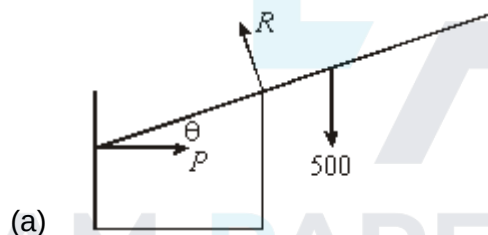
f.t. on magnitude

A1FA1

2

[10]

Q34.



All three forces in correct directions

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1

- (b) Resolving vertically

$$500 = R \cos \theta \quad \left(\Rightarrow R = \frac{500}{\cos \theta} \right)$$

For resolution

M1A1

Moments about A

$$500 \times \frac{5}{2} \cos \theta = R \times \frac{2}{\cos \theta}$$

For taking moments

M1

LHS

A1

RHS

A1

$$\Rightarrow 500 = \frac{4R}{5 \cos^2 \theta}$$

$$\therefore R = \frac{2500 \cos^2 \theta}{4}$$

$$\Rightarrow \frac{500}{\cos \theta} = \frac{2500 \cos^2 \theta}{4}$$

or other appropriate equating of values

M1 A1

$$\Rightarrow \cos^3 \theta = \frac{4}{5}$$

$$\Rightarrow \theta = 21.8^\circ$$

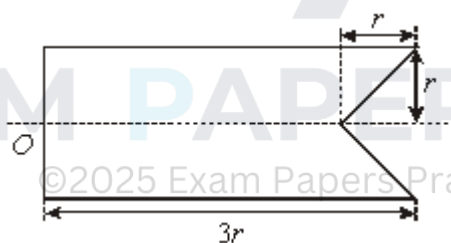
A1

8

[9]

Q35.

(a)



	Weight	Dist from O
Cylinder	$3\pi r^3 \rho$	$\frac{3r}{2}$
Cone	$\frac{1}{3} \pi r^3 \rho$	$2r + \frac{3r}{4} = \frac{11r}{4}$
New body	$\frac{8}{3} \pi r^3 \rho$	$\bar{x}$

3 shapes

M1

$$-1EE$$

A2

$$\frac{8}{3} \pi r^3 \rho \bar{x} + \frac{1}{3} \pi r^3 \rho \times \frac{11r}{4} = 3\pi r^3 \rho \times \frac{3r}{2}$$

correct combination

M1

at least 2 weight $\times$ dist correct

A1

$$\frac{8\bar{x}}{3} = \frac{9r}{2} - \frac{11r}{12}$$

$$\Rightarrow 32\bar{x} = 54r - 11r$$

$$\Rightarrow \bar{x} = \frac{43r}{32}$$

A1

6

(b)



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Body will topple when $\alpha > \tan^{-1} \frac{r}{\bar{x}}$

M1

$$\alpha > \frac{\tan^{-1} r}{\frac{43r}{32}}$$

A1f.t.

$$> \tan^{-1} \frac{32}{43}$$

$$> 36.7^\circ$$

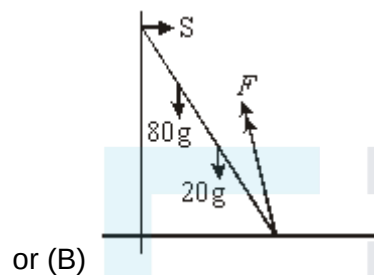
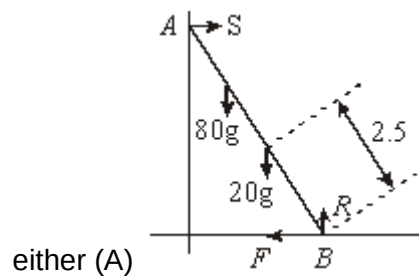
A1f.t.

3

[9]

Q36.

- (a)
(b) (i)



*-1 EE
if (B), F must be in approximately correct direction*

A2,1

2

Res vert $R = 80g + 20g = 100g$

B1

Res horiz $S = F$

B1

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and $F = 0.4R$ (limiting equilibrium)

Allow B1 if appears on diagram only

B1

$\Rightarrow S = 0.4 \times 100g$

B1

4

$= 392\text{N}$

AG

(or any other point)

- (ii) Moments about B

M1

$20g \times 2.5 \cos 60^\circ + 80g \times x \cos 60^\circ = S \times 5 \sin 60^\circ$

(–1 for each incorrect term)

A2,1

$$\Rightarrow 50g + 80gx = 40g \times 5 \tan 60^\circ$$

$$\Rightarrow 5 + 8x = 20 \tan 60^\circ$$

for correct progress

A1F

$$\Rightarrow x = \frac{20 \tan 60^\circ - 5}{8}$$

$$\approx 3.71 \text{ (3.5051)}$$

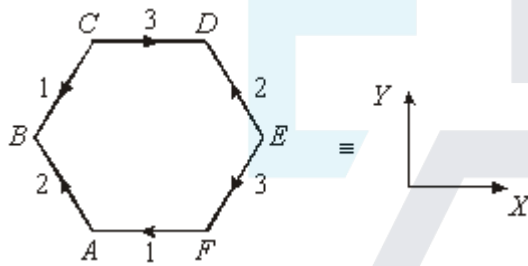
$$\text{AWRT; ft or } x = \frac{5(4\sqrt{3} - 1)}{8}$$

A1F

5

[11]

Q37.



(a) $X = -2\sin 30^\circ - 1\sin 30^\circ + 3 - 2\sin 30^\circ - 3\sin 30^\circ + 1$

M1

$$= 4 - 8\sin 30^\circ$$

A2,1

$$= 0$$

$$Y = 2\cos 30^\circ - 1\cos 30^\circ + 0 + 2\cos 30^\circ - 3\cos 30^\circ + 0$$

M1

$$= 0$$

A1

5

(b) (i) Taking moments about A

M1

$$L = 1 \times a\cos 30^\circ - 3 \times 2a\cos 30^\circ + 2 \times 2a\cos 30^\circ - 3 \times a\cos 30^\circ$$

A1

$$= -4a\cos 30^\circ$$

$$= -2\sqrt{3}a \quad (-3.46a)$$

$$\therefore \text{magnitude} = 2\sqrt{3}a$$

ft on 1 error

A1F

3

- (ii) couple acts clockwise
ft on answer to (b)(i)

A1F

1

[9]

Q38.

(a)
$$\bar{x} = \frac{3 \times 1 + 12 \times 2.5 + 15 \times 3 + 10 \times 2}{3 + 12 + 15 + 10}$$

Five term moment equation

Correct numerator

M1

A1

$$= \frac{98}{40} = 2.45 \text{ m}$$

Correct denominator

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A1

ag *Correct answer from correct working*

A1

4

(b)
$$4T_B = 2.45 \times 40 \times 9.8$$

Two term moment equation

M1

$$T_B = 240 \text{ N (to 3 sf)}$$

Correct moment equation

A1

Correct tension

A1

$$T_A + 240 = 40 \times 9.8$$

Vertical equilibrium or second moment equation

m1

$$T_A = 152 \text{ N (to 3 sf)}$$

Correct second tension

A1

5

[9]

Q39.

(a) (i) $\mathbf{F} = \{(8\mathbf{i} + \mathbf{j} - 2\mathbf{k}) + (-4\mathbf{i} + 2\mathbf{j} + 14\mathbf{k})\}$

M1

$$= -4\mathbf{i} - 3\mathbf{j} - 12\mathbf{k}$$

A1

2

(ii) Magnitude = $\sqrt{4^2 + 3^2 + 12^2}$
 $= 13$

M1

A1ft

2

(b) Moments about origin
(Use of $\mathbf{r} \times \mathbf{F}$)

M1

$$= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & -1 & 4 \\ 8 & 1 & -2 \end{vmatrix} + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 7 & -3 & 2 \\ -4 & 2 & 14 \end{vmatrix}$$

M1 for use of determinant or at least 2 terms correct in either moment

M1

$$= (-2\mathbf{i} + 36\mathbf{j} + 10\mathbf{k}) + (-46\mathbf{i} - 106\mathbf{j} + 2\mathbf{k})$$

(either moment)
(second moment)

A1

$$= -48\mathbf{i} - 70\mathbf{j} + 12\mathbf{k}$$

A1

5

Q40.

Resolve along AB

$$R \cos 30 = Q \cos 45$$

M1A1

Resolve perpendicular to AB

$$R \sin 30 + Q \sin 45 = P$$

M1A1

$$\frac{R}{2} + \frac{Q}{\sqrt{2}} = 4\sqrt{2}$$

$$\frac{R\sqrt{3}}{2} = \frac{Q}{\sqrt{2}}$$

*If just substitute given R into one of resolve eqⁿs,
M1 A1 only.*

$$\therefore \frac{R}{2} + \frac{R\sqrt{3}}{2} = 4\sqrt{2}$$

M1

$$R(1 + \sqrt{3}) = 8\sqrt{2}$$

$$R = \frac{8\sqrt{2}}{1 + \sqrt{3}} = \frac{8\sqrt{2}(\sqrt{3} - 1)}{2}$$

$$R = 4(\sqrt{6} - \sqrt{2})$$

A1

$$Q = \frac{\sqrt{3}}{\sqrt{2}} [4(\sqrt{6} - \sqrt{2})]$$

$$= 4(3 - \sqrt{3})$$

A1

7

[7]

Q41.

(a) (i)
$$\mathbf{F} = \begin{pmatrix} 1 \\ 3 \\ 4 \end{pmatrix} + \begin{pmatrix} 3 \\ -3 \\ 5 \end{pmatrix} + \begin{pmatrix} -1 \\ 4 \\ 3 \end{pmatrix}$$

M1

$$= \begin{pmatrix} 3 \\ 4 \\ 12 \end{pmatrix}$$

A1

2

(ii) Magnitude $= \sqrt{3^2 + 4^2 + 12^2}$

M1

$$= 13$$

A1

2

(b) (i) Displacement of force from (3, 8, 7)

$$\text{is } \begin{pmatrix} 5 \\ -4 \\ 7 \end{pmatrix} - \begin{pmatrix} 3 \\ 8 \\ 7 \end{pmatrix}$$

$$= \begin{pmatrix} 2 \\ -12 \\ 0 \end{pmatrix}$$

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B1

Moment is $\mathbf{r} \times \mathbf{F}$

=

$$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & -12 & 0 \\ -1 & 4 & 3 \end{vmatrix}$$

M1 if one row [and $\mathbf{i}, \mathbf{j}, \mathbf{k}$] correct

M1

A1

$$= (-36\mathbf{i} - 6\mathbf{j} - 4\mathbf{k})$$

sc 3 for $36\mathbf{i} + 6\mathbf{j} + 4\mathbf{k}$

A1

(ii) Moment of $\mathbf{F}$ is $-36\mathbf{i} - 6\mathbf{j} - 4\mathbf{k}$

B1✓

Moment of the other two forces is zero
since they pass through (3, 8, 7)

E1

2

[10]

Q42.

(a) $\mathbf{F} = (7\mathbf{i} + 2\mathbf{j}) + (-3\mathbf{i} + 4\mathbf{j}) + (\mathbf{i} + 6\mathbf{j})$

M1

$$= 5\mathbf{i} + 12\mathbf{j}$$

A1

Magnitude of $\mathbf{F}$ is $\sqrt{5^2 + 12^2}$

M1

$$= 13$$

A1

4

(b) Moments about O;

Can take moments about (x, 0)

$$12x$$

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M1 A1

$$= 3 \times 1 + 4 \times 4 + 6 \times 8 - 1 \times 2 + 2 \times 3 + 7 \times 5$$

M1 A1

$$12x = 106$$

$$x = \frac{106}{12} = \frac{53}{6}$$

A1

5

Point is $(\frac{53}{6}, 0)$

Can use printed result

[9]

Q43.

$$(a) \quad \mathbf{F} = - \left\{ \begin{pmatrix} 1 \\ -5 \\ 2 \end{pmatrix} + \begin{pmatrix} 2 \\ -7 \\ -6 \end{pmatrix} \right\}$$

M1

[− sign]

M1

$$= \begin{pmatrix} -3 \\ 12 \\ 4 \end{pmatrix}$$

A1

3

$$(b) \quad \text{Magnitude} = \sqrt{(-3)^2 + 12^2 + 4^2}$$

M1

$$= 13$$

A1ft

2

[5]

Q44.

R is (1, 5, 5)

M1

$$\mathbf{r} - \mathbf{p} = \begin{pmatrix} -2 \\ 1 \\ 4 \end{pmatrix}$$

A1

$$\text{Moment is } (\mathbf{r} - \mathbf{p}) \times \mathbf{F} = \begin{vmatrix} i & j & k \\ -2 & 1 & 4 \\ 7 & -5 & 2 \end{vmatrix}$$

M1A1

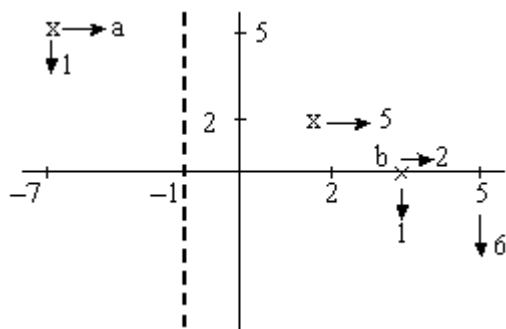
$$= 22\mathbf{i} + 32\mathbf{j} + 3\mathbf{k}$$

M2 A2 for $- [22\mathbf{i} + 32\mathbf{j} + 3\mathbf{k}]$

A1

[5]

Q45.



Resolve in x direction: $a + 5 + 2 = 0$

No marks for resultant $(7 + a)\mathbf{i} - 8\mathbf{j}$

M1

$$a = -7$$

A1

Taking moments about $(-1, 0)$:

M1

$$5 \times 2 + (1 + b) \times 1 + 6 \times 6 + a \times 5 - 1 \times 6 = 0$$

3 correct

A1

$$41 + b + 5a = 0$$

Rest correct
[About $(0,0)$ M1
33+b+5a A2]

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A1

$$41 + b - 35 = 0$$

M1

$$b = -6$$

A1

[7]

Q46.

(a) (i)

$$\underline{F} = - \left\{ \begin{pmatrix} 3 \\ 6 \\ -2 \end{pmatrix} + \begin{pmatrix} 2 \\ -2 \\ -1 \end{pmatrix} \right\}$$

If lose $-$ sign, SC1

M1

$$= -5\mathbf{i} - 4\mathbf{j} - 3\mathbf{k}$$

A1

2

(ii) Magnitude is $\sqrt{5^2 + 4^2 + 3^2}$

M1

$$= 5\sqrt{2}$$

oe

A1 ft.

2

(b) Moment of couple is moment of forces about O

M1

$$= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 4 & -2 & 1 \\ 3 & 6 & -2 \end{vmatrix} + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 5 & 3 & -4 \\ 2 & -2 & -1 \end{vmatrix}$$

$[\mathbf{r} \times \mathbf{F}]$

M1

$$= -2\mathbf{i} - 11\mathbf{j} + 30\mathbf{k} + -1\mathbf{i} - 3\mathbf{j} - 16\mathbf{k}$$

At least 5 terms correct

A1

$$= -13\mathbf{i} + 8\mathbf{j} + 14\mathbf{k}$$

cao

A1

5

[9]

Q47.

(a) **i** direction: $4 + 2 + a = c$

At least 3 correct

M1

j direction: $-3 + -1 + b = c$

At least 3 correct

M1

$$6 + a = c$$

A1

$$b - 4 = c$$

sc M1 for $F = (6 + a)\mathbf{i} + (b - 4)\mathbf{j}$

A1

Moments about O :

$$2 \times 3 + 4 \times 1 - 5 \times b + c \times a = -34$$

At least 3 correct

M1 A1

$$-5b + ac = -44$$

$$-5(4 + c) + (c - 6)c = -44$$

M1

$$c^2 - 11c + 24 = 0$$

$$(c - 8)(c - 3) = 0$$

$$c = 3, 8$$

A1

8

(b) $c = 3, a = -3, b = 7$

B1

$$c = 8, a = 2, b = 12$$

B1 for any 2 correct

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B1

2

[10]

Examiner reports

Q1.

This question was designed to discriminate between grades A and A*. In part (a) many students did not appreciate that the reason the reaction between the rod and the hemisphere acted through O was because the rod and hemisphere were both smooth. This fact was also needed to draw the force diagram in part (b) as at C the reaction was also perpendicular to possible movement. Thus the reaction at C acted perpendicular to the rod. However many students, when drawing the force diagram, did not show the reaction at point A acting through O . There was much inventive algebra shown in part (c) to obtain the printed result. Many students attempted to take moments about point A but were unable to find the length of AC in terms of l and a . The simplest method to find the length of AC was to draw the line from O to the midpoint of AC giving the length to be $2a \cos \theta$. However many used the cosine rule in triangle OAC and obtained $AC^2 = 2a^2 - 2a^2 \cos(180 - 2\theta)$ which was transformed into $AC^2 = 2a^2 + 2a^2 \cos 2\theta$. Very few students could transform this into $AC^2 = 4a^2 \cos^2 \theta$ and hence into $AC = 2a \cos \theta$. As expected only the better students completed this question successfully.

Q2.

This question was well answered. Almost all students chose to find the limiting P value for each case and then set up the inequality by arguing which value of P would be reached first. To score full marks, students had to clearly state why the inequality was needed by referring to sliding occurring before toppling. On occasions the inequality was reversed and not corrected. A more significant error was to use the mass of 0.4 kg when taking moments, leaving out g .

Q3.

This topic was done well, with (c) proving to be the most challenging. Part (a) was almost 100% correct. When errors occurred in part (b) they were largely due to sign errors although a handful of students attempted $\mathbf{F} \times \mathbf{r}$. Best solutions seen in part (c) adopted a step by step approach which was easy to mark. Less worthy attempts included not reversing signs of the force of moment from earlier parts, using the magnitude of forces rather than the force itself and failing to appreciate that the coordinates of a point on the y axis means that $x = z = 0$. Notation occasionally lost marks with vector forces not being identified as such.

Q4.

Most candidates completed the force diagram correctly; a few showed their frictional force acting away from the wall. In part (b), most candidates appreciated that taking moments was needed, however a few forgot to include some distances or angles. Part (c) was well answered by most candidates, regardless of whether they were able to answer part (b), as it was quickly noticed that F was 256 N (given in part (b)) and R was 100g (resolving vertically).

Q5.

This question was well attempted. There were fewer marks lost through careless errors with the determinant of the use of $\mathbf{F} \times \mathbf{r}$. Part (c) proved little trouble for most candidates,

although on occasions squaring went wrong or the solution of a quadratic eluded some candidates.

Q6.

The early part of this question was well answered. Part (b)(ii) proved most challenging with candidates mixing up clockwise and anticlockwise moments. A few candidates did not equate the individual sum of moments to the equivalent moment of -24 Nm . Poorest answers came from candidates who tried to apply $r \times F$ rather than simply use force $\times$ distance on a diagram. Part (c) was not well understood by a significant number of candidates.

Q7.

Candidate responses were more mixed than expected. A number of errors that lost accuracy marks occurred in part (b): for example, a wrong sign in the $\mathbf{j}$ component, simple numerical errors such as $0 \times 2 = 2$ or loss of negative signs elsewhere. Candidates who used $\mathbf{F} \times \mathbf{r}$ were penalised only the final mark if the solution was otherwise fully correct. For part (c), candidates had to make reference to the moment calculated in part (b) and the resultant force being zero.

Q8.

This is one topic that candidates do not appear to be confident with. In part (a), sign or magnitude mismatches occurred with regularity. A small number of candidates left the requested magnitude as a negative answer. Candidates were required in some way to comment on a cancelling or point to it explicitly to be awarded full marks. Vector product methods were done well. Part (b)(i) was done better than in the past, although some left the answer with a in it. Candidates would have been well advised to sketch a simple diagram, as in the mark scheme, for part (b)(ii) as it would lessen the error of ' $8d$ ' occurring.

Q9.

Many candidates showed an incorrect force diagram in part (a). Often no friction at C was shown, whereas some included friction at A . In the required moment equations, distance or angle considerations were often omitted. Candidates were more successful in part (d), usually gaining follow through marks.

Q10.

This was a question in which candidates scored very well indeed. In part (a), using $P \sin \theta$ was a common error. Almost all candidates obtained full marks on part (b). In part (c), the inequality was sometimes reversed. Candidates' reasoning in part (d) was excellent.

Q11.

Candidates did well with this unusual question. The printed answer in part (a) was successfully obtained by many candidates. Some made the usual error of $\mathbf{F} \times \mathbf{r}$ and then adjusted their answer to match the printed one. Part (b)(i) was done well too, with candidates sometimes showing that $a = 7$ gave the right magnitude and then finding the other correct value. Where candidates formed a quadratic equation, it was solved very efficiently. The last part discriminated well, and the most able candidates were able to see how to obtain the correct $\mathbf{F}$.

Q12.

The majority of candidates found the normal reaction at the foot of the ladder to be $112g$ and readily found the frictional force, $\mu \times 112g$, as given. In part (b), most appreciated that moments were needed and tried to take moments about A . However, in the terms, many forgot the distance or the angle necessary in taking moments.

Q13.

Candidates were very successful at part (a). Several types of errors occurred in part (b):

- $\mathbf{F} \times \mathbf{r}$ rather than $\mathbf{r} \times \mathbf{F}$;
- use of the wrong $\mathbf{r}$ — either the negative of the correct $\mathbf{r}$ or the coordinates of P ;
- expanding the correct determinant but omitting a minus sign.

Many candidates gave good explanations for part (b)(ii) — candidates who dropped marks in part (b)(i) could recover here on a follow-through principle.

Q14.

Part (a) was the part most successfully answered, though candidates must be aware that it is necessary to show full working when a printed answer is given. In part (b), the principles of moments were understood though not always applied with full success. Common errors were:

- inconsistent directions;
- failure to understand what to do with the couple of magnitude 20 Nm ;
- use of the magnitude of the resultant rather than the required x -component.

It was disappointing to see very few candidates attempt a vector equation. When it was attempted, quite often only the right-hand side was given.

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Q15.

Although a few candidates showed the frictional force acting in the opposite direction to the correct one, in general part (a) was answered well. Part (b) was usually correct, but major problems occurred in part (c). The correct use of the angles required in the calculations of the moments was often missing.

Q16.

Marks were rarely dropped, although some sign errors did occur. There was a great deal of variation in the point about which to take moments. Candidates who chose point C tended to do better. A small number of candidates chose a point halfway down the rod, which meant that three individual moments were combined, creating extra work.

Q17.

Responses to this question were more varied. Candidates often lost a mark in part (b) through using $F \times r$ or by making an error with the determinant. In part (c), candidates must appreciate that, when answers are given, full working must be shown. There was mixed success with part (d), with either the vector product or scalar product method being

used. Sometimes these were mixed together. Other errors consisted of using the vectors from the wrong triangle, eg OA and OB .

Q18.

Part (a) was done well although not all candidates related their coordinates of the centre of mass to the line CH . Part (b) was mostly done well but those taking moments about a point other than the foot of the lamina sometimes omitted a force. Candidates knew the principle of mechanics required in answering part (c). However, weaknesses in handling inequalities spoiled many otherwise good solutions.

Q19.

Topic – Couples in 3D. There was a very good response to this question. Most candidates knew that the sum of forces must be zero and used it to find $\mathbf{F}$, clearly showing their reasoning.

The most common error was to add the given two forces and give this as the answer.

Showing that the magnitude of $\mathbf{F}$ was $3\sqrt{10}$ proved straightforward, although $\sqrt{90}$ had to be seen to secure both marks. Almost all candidates correctly used $\mathbf{r} \times \mathbf{F}$ although a common error was not to use all the necessary determinants. The most efficient solutions found $\overline{QP}$ and then used just one determinant.

Q20.

Topic – Toppling and sliding. There was an excellent response to this question. Part (a) was almost universally correct; however when slips did occur it was because of the use of $W(4a)$ when taking moments.

Part (b) was rarely seen incorrect, however inequality signs sometimes lost a mark. Part (c) was answered very well with sound knowledge of cancelling, and trigonometrical identities. Once again a mark was sometimes lost because of an equality or greater than sign.

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Q21.

Most candidates knew that the result must be zero and used it to find (a) and (b). In part (b) minor slips in the calculation of the determinant lost candidates marks. The best solutions showed clear calculation of each determinant before adding them together. Calculation of $\mathbf{F} \times \mathbf{r}$ resulted in a 1 mark penalty.

Q22.

Part (a) was very well done. In part (b) some candidates again used $\mathbf{r} \times \mathbf{F}$, although when candidates chose to do this they often made errors. The most successful candidates calculated the moments from a clear diagram. Candidates did clearly distinguish between clockwise and anticlockwise moments. In part (b) the most common error was to use the total resultant instead of the x component. A number of correct solutions were seen which used the equation of the line of action.

Q23.

This proved to be popular and many candidates obtained full marks. Part (a) was often done well, but some omitted one of the forces at the foot of the ladder while others

showed reaction forces perpendicular to the ladder. Part (b) was mostly successful, and there were many highly competent attempts at the moments equation in part (c), with the most common errors being trigonometrical or slips in subsequent working.

Q24.

Part (a)(i) proved straightforward for the majority of candidates. Errors in parts (a)(ii) and (iii) usually resulted from not taking the double thickness of the triangular section into consideration. In part (b) the idea of the lamina being held in equilibrium by the action of a couple proved difficult for many.

Q25.

All candidates were able to calculate the equivalent force in part (a). Candidates generally show a preference for working with column vectors, rather than unit vectors, and reformat data given in the latter form prior to performing calculations. Again, it is recommended that for clarity in these questions, a second “Equivalent” diagram is drawn and that equivalent forces (or couples) not be superimposed on the diagram showing the basic data. Also, candidates are warned against working out answers which are not required – in this case the equation of the line of action of the equivalent force. This gains no credit and merely uses up valuable time.

Q26.

This question was competently answered by many, with candidates demonstrating their confidence in handling inequalities. By contrast however, there still remains lack of confidence and efficiency in calculation when dealing with situations where the given data includes such as $\tan \theta = 3$. Taking moments about the base of the ladder results in an equation involving one multiple of $\sin \theta$ and two of $\cos \theta$. Division by $\cos \theta$ results in one multiple of $\tan \theta$ for which the value, 3, may be substituted. Many instances were seen, even on otherwise good scripts, where $\sin \theta$ and $\cos \theta$ were calculated and inserted in the equation representing the moments.

Q27.

Most candidates were clear as to the tension in the rope and the position and direction of the weight of the beam in part (a), but the direction of the reaction on the beam at the wall proved beyond most. In part (b) only a minority of candidates were able to handle the inequality with confidence and clarity. Most of the remainder were able to calculate the angle, but the reasons as to why this should be greater than 53° were generally lacking.

Q28.

Most candidates answered part (a) competently. Calculating the moment of each force about the origin proved difficult in part (b) for weaker candidates. The question asked for the magnitude of the resultant moment to be stated specifically. An appreciable number of candidates omitted to do so. Candidates who had answered the previous part correctly generally had no difficulties with part (c), although there were some exceptions. A few seconds spent sketching a diagram of the equivalent force acting at the point (0, d) would probably have increased the success rate greatly.

Q29.

Generally this was the most successfully attempted question. Some candidates omitted the weight of the ladder from their calculation. Care was necessary in taking moments,

about whatever chosen point (axis). Given the angle $\tan^{-1}12/5$, it was not expected that candidates would calculate the size of the angle and then subsequently use calculators when substituting for sin and cos. Candidates were expected to recognise this basic Pythagorean triad.

Q30.

It was anticipated that the majority of candidates would attempt to resolve forces parallel to each axis in part (a) and attempt to take moments about the origin in part (b)(i), thus at least gaining some method marks, but a high proportion of candidates did not appear able to tackle this. Some candidates, otherwise in difficulties with part (b)(i), were able to gain follow through marks in finding a solution to part (b)(ii).

Q31.

This question required clarity of thought. Most candidates were able to make some progress. The use of the word “toppling”, as in the specification, led some candidates mistakenly to draw a diagram with the block rotated through 45° .

Q32.

The first parts were generally well answered. Weaker candidates calculated the direction of the force at the hinge, rather than the angle made by the rod with the vertical.

Q33.

The first part, to calculate the magnitude of the resultant force, was well answered but the moments calculation in the second proved more difficult. Some candidates would have benefited from drawing a second diagram for clarity when calculating values of a system equivalent to the given one. Candidates who also gave unrequired extras, such as the direction of the line of action of the force, or its equation, did not gain extra credit and wasted time.

Q34.

Most candidates had initial difficulties with determining the directions of forces at smooth contacts.

Q35.

This was a situation where the final body was obtained by subtracting the cone from the cylinder, and this caused candidates difficulties in drawing up an appropriate “moments equation”. With the answer to the first part having been given, the solution to the final “toppling” problem was standard of its type. Candidates should be aware that they are not penalised here if they have not satisfactorily answered the first part of such a question, but go on to use the information given to pick up later marks.

Q36.

This question appeared to pose no difficulties for the majority of candidates.

Q37.

Most candidates were able to resolve the forces in two mutually perpendicular directions to show the zero resultant in part (a). The more confident ones wrote this down as a single vector statement. In part (b), the successful candidates were clear about which sign

convention (anticlockwise positive, for example) they were using, although the basic geometry of the hexagon gave problems to some.

Q38.

This question caused more difficulty for a fair number of candidates. Quite a lot managed to produce correctly the printed answer for part (a).

Part (b) caused much more difficulty. Some candidates did not attempt this or did not get beyond a statement like $T_1 + T_2 = 392$. Some did produce quite a bit of incorrect working. There were some candidates who did not multiply the unknown tension by a distance, although they did do this for the weight of each particle and of the rod. Interestingly quite a few candidates did not use the answer from part (a), but treated each particle separately. Another common error was to omit g from the calculations.

Q39.

Part (a)(i) was completed well with only a few candidates obtaining sign errors. Part (a)(ii) was completed well by all candidates.

Most candidates were also successful in part (b) with only a few arithmetic errors being seen.

Q40.

Most candidates answered this question well. A few assumed that $R = 4(\sqrt{6} - \sqrt{2})$, and substituted into one equation, either obtained by resolving along or perpendicular to AC , to find Q .

Q41.

Part (a) was answered well by all candidates. In part (b), many candidates did not find the distance of the line of action of the force from the point $(3, 8, 7)$. Amongst those who did, sign errors were common in the resulting moment calculation. Most candidates appreciated that the moment of $\mathbf{F}$ was the same as the moment of the force

$$\begin{pmatrix} -1 \\ 4 \\ 3 \end{pmatrix}$$

because the other two forces had their line of action through $(3, 8, 7)$.

Q42.

This question was answered well, with almost all candidates finding the force $\mathbf{F}$ to be $5\mathbf{i} + 12\mathbf{j}$, but a minority forgot to find the **magnitude** of the force. In part (b), nearly every candidate took moments about 0 and the majority found 106 (or -106) as the total moment and hence obtained the printed result.

Q43.

This question was answered well. A few candidates found the sum of the two forces rather than the force needed to form a couple.

Q44.

Candidates attempted to find the mid-point of PQ or find PR. Many errors in signs were seen, together with numerical errors. To find the moment of F, it was relatively rare to

see $\begin{pmatrix} -2 \\ 1 \\ 4 \end{pmatrix} \times \begin{pmatrix} 7 \\ -5 \\ 2 \end{pmatrix}$ attempted.

Q45.

Most candidates correctly obtained $(7 + a)\mathbf{i} - 8\mathbf{j}$ as the resultant of the given forces.

Unfortunately most did not then appreciate that for the resultant to act along $x = -1$, the i component had to be zero. Thus, when they took moments about either $(-1, 0)$ or $(0, 0)$ the candidates only had one equation in two unknowns, which could not be solved.

Q46.

In part (a), a significant proportion of candidates found the resultant of the two forces rather than the force needed to form a couple. Part (b) was well answered. In part (c), a number used $\mathbf{F} \times \mathbf{r}$ rather than $\mathbf{r} \times \mathbf{F}$ to find the moment.

Q47.

Many candidates found the resultant to be $(6 + a)\mathbf{i} + (b - 4)\mathbf{j}$ but rarely equated this to $c\mathbf{i} + c\mathbf{j}$. Hence, after taking moments, they obtained just one equation: $-5b + ac = -44$ in three unknowns which they could not solve. There were many errors in the signs of the terms obtained by taking moments about O . This was due in part to few candidates using clear diagrams. Those who answered part (a) correctly usually had no difficulty with part (b).